When Debiasing Hurts: Measurement Error Amplification in LLM-Annotated Heterogeneous Treatment Effects

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Abstract

LLM annotations are increasingly used as effect modifiers in heterogeneous treatment effect (HTE) estimation for computational social science. We identify a systematic failure mode: when LLM misclassification rates vary across subgroups, applying a global (pooled) confusion matrix correction to subgroup-level CATE estimates introduces bias proportional to the product of misclassification heterogeneity and CATE heterogeneity ($\Delta\pi \times \Delta\tau$). Through a large-scale empirical study spanning 3,600 synthetic Monte Carlo runs, semi-synthetic CSS benchmarks, real-data evaluation on hate speech corpora, and 6 LLM annotators across 3 model generations, we characterize when this amplification occurs and how severe it becomes. Global correction worsens subgroup CATE bias by 2–3$\times$ (median) and up to 5.9$\times$ with nonlinear estimators. We attempt correction via hierarchical Bayesian stratified estimation (EC-HTE), but find it fails on real data: on the UC Berkeley Hate Speech corpus, EC-HTE produces *higher* bias than both naive estimation and global correction. A proposed diagnostic criterion achieves AUC of only 0.49 at the actionable setting level — no better than random. Our results demonstrate that standard debiasing pipelines can systematically harm downstream causal estimates, and that correcting this failure mode remains an open problem.

1 Introduction

Large language models are rapidly replacing human annotators in computational social science (CSS) research (Tornberg, 2024; Gilardi et al., 2023; Ziems et al., 2024). When LLM annotations serve as *effect modifiers* — variables that define subgroups for heterogeneous treatment effect (HTE) analysis — misclassification introduces measurement error into the causal estimation pipeline. A growing body of work addresses this through prediction-powered inference (Angelopoulos et al., 2023a), doubly-robust subpopulation estimation (Egami et al., 2024), and outcome-side correction (Sui and Egami, 2026). These methods typically apply a *global* confusion matrix estimated from a small expert-labeled calibration set.

We identify a failure mode that arises precisely when the effect modifier itself is mismeasured: LLM misclassification rates are not uniform across subgroups. When a pooled confusion matrix is applied to correct subgroup-level CATEs, it systematically over-corrects low-error subgroups and under-corrects high-error ones. The resulting bias is proportional to $\Delta\pi \times \Delta\tau$ — the product of misclassification heterogeneity and treatment effect heterogeneity. This relationship is a special case of classical measurement error theory (Carroll et al., 2006; Fuller, 1987), but its practical consequences for LLM annotation pipelines have not been characterized.

We present a large-scale empirical study documenting this phenomenon across synthetic, semi-synthetic, and real-data settings. Our findings are sobering:

- **Global correction amplifies bias** in $\sim$50% of the parameter space, by 2–3$\times$ (median) and up to 5.9$\times$ with CausalForest estimators (§5.1, §5.9).
- **Proposed corrections fail on real data**: hierarchical Bayesian stratified correction (EC-HTE) produces the *worst* bias among all methods on the UC Berkeley Hate Speech corpus (§5.4).
- **No reliable diagnostic exists**: a criterion based on $\Delta\pi \times \Delta\tau$ achieves setting-level AUC of 0.49 — worse than a coin flip (§5.5).
- **Heterogeneity is structural**: across 6 LLMs spanning 3 model generations, misclassifica-

tion heterogeneity ($\Delta\pi = 0.08$–0.20) persists even in the latest models (§5.8).

Our contribution is primarily empirical: we characterize a practically important failure mode, document the conditions under which it occurs, and show that naïve corrections — including our own proposed method — do not reliably solve it. We hope this serves as a warning to CSS researchers using LLM annotations in causal pipelines and motivates future methodological work.

2 Related Work

LLM-as-Annotator. LLMs achieve near-human annotation quality across CSS tasks including sentiment analysis, stance detection, and toxicity classification (Gilardi et al., 2023; Tornberg, 2024; Ziems et al., 2024). However, error rates vary systematically across demographic and linguistic subgroups (Ashwin et al., 2025), creating the heterogeneous misclassification patterns we study.

Prediction-Powered Inference. PPI (Angelopoulos et al., 2023a) and its extensions (Angelopoulos et al., 2023b) use a small labeled dataset to correct ML predictions for valid statistical inference. DSL (Egami et al., 2024) extends this to causal estimands in social science. PPCI (Sui and Egami, 2026) handles conditional estimands including CATEs. All apply global corrections that assume homogeneous prediction error — the assumption we show fails.

Measurement Error in Causal Inference. Classical measurement error theory (Fuller, 1987; Carroll et al., 2006; Gustafson, 2003) establishes that misspecified corrections can amplify MSE. Imai and Yamamoto (2010) study measurement error in causal mediation. Mozer (2026) proves arm-specific rectifiers fix ATE under differential prediction error, but limits analysis to average effects. ForestIV (Bu et al., 2023) handles ML-covariate measurement error in regression but not CATE estimation.

Heterogeneous Treatment Effects. Honest causal forests (Wager and Athey, 2018; Athey et al., 2019), doubly-robust learners (Kennedy, 2023), and meta-learners (Chernozhukov et al., 2018) estimate CATEs from observational data. None address the case where the variable defining treatment effect heterogeneity is itself mismeasured by an ML system.

Bayesian Annotation Models. Dawid–Skene models (Dawid and Skene, 1979) and their extensions (Passonneau and Carpenter, 2014; Hovy et al., 2013) estimate annotator-specific confusion matrices. Our hierarchical Bayesian formulation is related but targets confusion matrix estimation for causal correction rather than label aggregation.

3 Bias Mechanism

3.1 Setup and Notation

Let $Z \in \{0, 1\}$ denote a binary effect modifier (e.g., hate speech type), $\hat{Z}$ its LLM prediction, $W \in \{0, 1\}$ a binary treatment, and Y the outcome. We partition the population into K subgroups $\{S_k\}_{k=1}^{K}$ defined by covariates X. The subgroup-specific CATE is $\tau_k(z) = \mathbb{E}[Y(1) - Y(0) \mid Z = z, S_k]$.

Each subgroup S_k has a confusion matrix C_k with misclassification rates $\pi_k = P(\hat{Z} \neq Z \mid S_k)$. A *global* confusion matrix $\bar{C}$ pools across subgroups.

3.2 Bias Decomposition

The bias from applying global correction $\bar{C}^{-1}$ to subgroup-level CATE estimates decomposes as:

$$\text{Bias}_k = f(\Delta\pi_k, \Delta\tau_k) \tag{1}$$

where $\Delta\pi_k = \pi_k - \bar{\pi}$ is the subgroup's deviation from the global misclassification rate and $\Delta\tau_k$ captures CATE heterogeneity. The bias is proportional to the product $\Delta\pi \times \Delta\tau$.

This is a special case of the bias amplification results in Carroll et al. (2006) (Ch. 4) for misspecified measurement error corrections. We verify empirically that this identity holds in the LLM annotation context with $R^2 > 0.97$ across configurations (§5.1).

When debiasing hurts. When $\Delta\pi_k$ and $\Delta\tau_k$ have appropriate sign combinations, global correction *increases* bias relative to the naive (uncorrected) estimator. This occurs in approximately 50% of the parameter space. The maximum amplification reaches 2.86× for K=2 and 2.41× for K=4 with linear estimators, and up to 5.9× with CausalForest.

3.3 EC-HTE: Attempted Correction

We propose Error-Corrected Heterogeneous Treatment Effects (EC-HTE), a hierarchical Bayesian approach that estimates subgroup-specific confusion matrices $\hat{C}_k$ rather than applying the global $\bar{C}$.

For each subgroup k, we place Dirichlet priors on the rows of C_k:

$$C_k[z, \cdot] \sim \text{Dir}(\alpha_{z,1}, \ldots, \alpha_{z,J}) \quad (2)$$

with a shared hyperprior across subgroups that enables partial pooling (James–Stein shrinkage). The corrected CATE uses $\hat{C}_k^{-1}$ applied to the LLM-based estimate within each subgroup, requiring $\det(\hat{C}_k) \neq 0$.

PPCI composition. EC-HTE addresses covariate-side measurement error. When outcome-side error is also present, we compose EC-HTE with PPCI (Sui and Egami, 2026): EC-HTE handles structured covariate error via mixing matrix inversion, while PPCI's residual rectifier captures outcome error.

3.4 Diagnostic Criterion

We propose using $|\Delta\pi \times \Delta\tau|$ as a diagnostic to predict when global correction is harmful. Subgroups where this product exceeds a threshold should use stratified rather than global correction.

4 Experimental Setup

Synthetic DGP. We sweep $K \in \{2, 4\}$ subgroups, misclassification regimes (moderate: $\Delta\pi \leq 0.05$; extreme: $\Delta\pi \approx 0.12$), and expert budgets $n \in \{50, 100, 250, 500, 1000, 2000\}$ across 3,600 Monte Carlo runs. MCMC diagnostics confirm convergence ($\hat{R}_{\max} = 1.024$, $\text{ESS}_{\min} = 129$).

Semi-synthetic DGP. We use TweetEval sentiment data with injected CATE structure, 4 subgroups with controlled misclassification rates (S_0=0.05, S_1=0.10, S_2=0.15, S_3=0.25), and $P(Z = 1) = 0.39$.

Real data. UC Berkeley Measuring Hate Speech corpus (n=2,496 comments) with subgroup-varying LLM misclassification ($\Delta\pi = 0.117$, global FNR = 53%).

Baselines. Naive (no correction), Global (pooled CM correction), EC-HTE (proposed), PPCI (Sui and Egami, 2026), DSL per-subgroup (Egami et al., 2024).

K	R^2	Tightness	Note
2	0.9939	1.003	Exact match
4	0.9738	1.023	Finite-sample noise

Table 1: Bias identity verification. The product-form relationship Bias $\propto \Delta\pi \times \Delta\tau$ holds with $R^2 > 0.97$, consistent with classical measurement error theory (Carroll et al., 2006).

Scenario	Bias Ratio	$\Delta\pi$ Threshold
K=2 / extreme	2.86×	0.124
K=4 / extreme	2.41×	0.124
K=2 / moderate	1.20×	0.124

Table 2: Debiasing-hurts regions: maximum ratio of global-corrected bias to naive bias. Global correction can nearly triple bias in extreme heterogeneity settings.

5 Results

5.1 Bias Identity Verification

We first verify the classical bias decomposition (Eq. 1) holds empirically in the LLM annotation context. Table 1 confirms the product-form relationship with $R^2 > 0.97$ for both K=2 and K=4. The tightness ratios near 1.0 confirm this is an exact identity, not a bound.

Global correction amplifies bias in regions where $\Delta\pi$ and $\Delta\tau$ interact adversely. Table 2 quantifies the amplification. The maximum bias ratio reaches 2.86× for K=2 (extreme heterogeneity) — debiasing nearly triples the bias compared to leaving it uncorrected.

5.2 Synthetic Validation of EC-HTE

Under controlled synthetic conditions, EC-HTE's hierarchical Bayesian estimation provides modest RMSE improvements over MLE through partial pooling (Table 3). These gains (4–9% for K=2, n=50) reflect standard James–Stein shrinkage and diminish rapidly with sample size.

HB coverage ranges from 0.87 to 0.93 (median 0.93), below nominal 0.95 in the most challenging configurations. Hyperprior sensitivity is moderate (∼4–5% RMSE variation under 2× changes in Gamma hyperparameters for K=4).

5.3 Semi-Synthetic CSS Benchmark

On the TweetEval-based semi-synthetic benchmark (Table 4), EC-HTE shows a complex performance profile. At n=100 with extreme heterogeneity (S_3, misclassification rate 0.25), HB achieves 60% RMSE improvement over MLE.

Config	HB	MLE	Δ%
K=2 / extreme / n=50	0.091	0.095	−4.2
K=2 / extreme / n=100	0.079	0.080	−1.3
K=4 / extreme / n=50	0.99	1.28	−22.7
K=4 / moderate / n=50	0.14	0.15	−6.7

Table 3: HB vs. MLE RMSE on synthetic data. Improvements are modest for K=2 (standard partial pooling benefit) and larger for K=4/n=50 where per-subgroup estimates are noisiest.

Config	ΔRMSE	Practical?
n=50 / S_3 / extreme	−57%	No
n=100 / S_3 / extreme	+60%	Marginal
n=250 / S_3	+5–17%	Yes
n=500 / all	+0.2–1.6%	Redundant

Table 4: Semi-synthetic results (HB vs. MLE RMSE change). EC-HTE helps at n=100+ with extreme heterogeneity but *hurts* at practical budgets (n=50). This benchmark has $P(Z{=}1) = 0.39$; real datasets typically have lower positive rates.

However, at the more practical budget of n=50, HB is 57% *worse* — the per-subgroup confusion matrix estimation noise overwhelms the bias correction benefit. At $n \geq 500$, all methods converge and gains are negligible (0.2–1.6%).

Critically, this benchmark was designed with $P(Z{=}1) = 0.39$ and symmetric misclassification — conditions more favorable than typical real-world settings. The real-data evaluation (§5.4) uses $P(Z{=}1) = 0.18$ and FNR = 53%, where per-subgroup positive-class expert labels are scarce.

5.4 Real-Data Evaluation: The Negative Result

Table 5 presents the central result of this paper. On the UC Berkeley Hate Speech corpus with real LLM annotations, EC-HTE produces the *highest* average absolute bias among all methods, including doing nothing.

The failure is driven by two interacting factors. First, global FNR = 53% means that for the minority class ($P(Z{=}1) = 0.18$), per-subgroup positive-class expert labels are extremely sparse, making per-subgroup confusion matrix estimation unreliable. Second, $\Delta\pi = 0.117$ places the dataset in the *moderate* heterogeneity regime — precisely where our budget analysis (§5.6) shows EC-HTE never outperforms global correction.

This negative result is corroborated across multiple LLM annotation sources (Table 6).

Method	Avg |Bias|	Rank
Naive (no correction)	0.210	1 (best)
Global corrected	0.273	2
EC-HTE (proposed)	0.313	3 (worst)

Table 5: Real-data results on UC Berkeley Hate Speech (n=2,496). EC-HTE produces the worst bias. Doing nothing outperforms both correction methods. This pattern is confirmed by exp-015 (GPT-4.1 CMs: HB 3–4× worse) and exp-016 (GPT-3.5 CMs: HB 3.5–5× worse).

Experiment	CM Source	HB vs. Global	Verdict
exp-004	UC Berkeley	Worse	Negative
exp-015	GPT-4.1 pilot	3–4× worse	Negative
exp-016	GPT-3.5	3.5–5× worse	Negative

Table 6: Cross-validation of real-data failure across confusion matrix sources. The pattern is systematic, not a dataset artifact.

5.5 Diagnostic Criterion: A Negative Result

Table 7 evaluates the proposed diagnostic $|\Delta\pi \times \Delta\tau|$ for predicting when global correction is harmful. At the subgroup level, AUC reaches 0.833 — but this requires knowledge of true subgroup CATEs, which are unknown at decision time. At the *setting level*, where practitioners must decide which correction strategy to apply for a given dataset, AUC drops to 0.49: no better than random.

We also find that 84.6% of low-$\Delta\pi$ settings are hurt by correction — but this is a marginal distribution, not a prediction. The diagnostic adds no information beyond what the bias identity itself implies.

5.6 Expert Label Budget Analysis

Table 8 reports crossover points where EC-HTE begins to outperform global correction in RMSE. These budgets are impractical in the regime where most real data falls.

In the moderate-heterogeneity regime that characterizes real data ($\Delta\pi \approx 0.10$–0.12), EC-HTE either never outperforms global correction (K=2) or requires expert budgets (2,000 labels for K=4) that render the LLM annotation cost advantage moot. HB outperforms MLE in 21/24 (87.5%) within-method comparisons, confirming that partial pooling helps — but improving upon the wrong baseline does not change the fundamental conclusion.

Level	AUC	Actionable?
Subgroup	0.833	No (requires true CATE)
Setting	0.490	Yes, but fails
Asymmetric	0.611–0.626	No

Table 7: Diagnostic AUC at different granularities. The diagnostic is useless at the actionable level (AUC = 0.49). Subgroup-level AUC is high but circular — it requires the quantity being estimated.

Config	Crossover n	Practical?
K=2 / extreme	250	Marginal
K=2 / moderate	Never	No
K=4 / extreme	1,000	No
K=4 / moderate	2,000	No

Table 8: Expert label budget crossover points. In the moderate regime (common in real data), EC-HTE either never helps (K=2) or requires prohibitive budgets (K=4). At n=2,000, expert labels alone suffice for CATE estimation.

5.7 PPCI and DSL Baselines

Table 9 evaluates whether composing EC-HTE with PPCI adds value beyond PPCI alone in dual-error scenarios (both covariate and outcome measurement error).

PPCI alone achieves |Bias| = 0.004, matching the EC-HTE+PPCI composition (0.005). The composition provides a 14% RMSE improvement (0.225 vs. 0.261), but this is modest relative to the additional implementation complexity. EC-HTE alone (|Bias| = 0.178) cannot handle outcome error, confirming that PPCI is the more general-purpose correction.

5.8 Misclassification Heterogeneity Across LLMs

Table 10 characterizes misclassification heterogeneity across 6 LLMs spanning 3 model generations. All models exhibit $\Delta\pi > 0.05$, confirming that heterogeneous misclassification is a structural phenomenon, not a model-specific artifact.

Heterogeneity decreases across generations (GPT-3.5: 0.198 $\rightarrow$ GPT-5.4: 0.114) but remains in the moderate regime where our budget analysis shows EC-HTE provides little benefit. The worst-performing subgroup (True_medium) is consistent across 5/6 models, suggesting systematic difficulty with intermediate-severity content.

Method	|Bias|	RMSE
Naive	0.251	0.269
EC-HTE alone	0.178	0.226
PPCI alone	0.004	0.261
EC-HTE + PPCI	0.005	0.225

Table 9: DGP C (both errors): PPCI alone achieves nearly identical bias (0.004 vs. 0.005) to the composition. EC-HTE's marginal contribution is a 14% RMSE reduction — modest given the added complexity.

Model	Generation	$\Delta\pi$
GPT-3.5-turbo	Gen 1	0.198
GPT-4-turbo	Gen 2	0.149
GPT-4.1	Gen 2	0.098
GPT-4o	Gen 2	0.080
GPT-4o-mini	Gen 2	0.136
GPT-5.4	Gen 3	0.114

Table 10: Effective misclassification heterogeneity $\Delta\pi$ across 6 LLMs. Heterogeneity persists ($\Delta\pi = 0.08$–0.20) even in latest-generation models. 5/6 models share the same worst subgroup (True_medium).

5.9 CausalForest Integration

When EC-HTE is combined with CausalForestDML, the bias–variance tradeoff becomes particularly unfavorable (Table 11).

EC-HTE achieves the lowest mean absolute bias (0.061 vs. 0.107 for global), but at the cost of 38% higher RMSE (0.229 vs. 0.166). For practitioners who optimize prediction accuracy rather than unbiasedness alone, this tradeoff is unfavorable.

CausalForest also amplifies the debiasing-hurts effect: the maximum global/naive bias ratio reaches 5.9$\times$ in K=4/extreme configurations (Table 12), compared to 2.41$\times$ with the difference-in-means estimator. However, this extreme amplification occurs in only 9/144 (6.25%) of configurations.

5.10 Robustness to Subgroup Misspecification

Table 13 evaluates EC-HTE when subgroup assignments contain noise (0–50% random label flips). Coverage remains near nominal (0.945–0.958) across all noise levels, and HB outperforms MLE in every configuration.

This graceful degradation is expected: as subgroup labels become random, the hierarchical posterior concentrates on the global mean, recovering global correction by construction. This is a standard property of Bayesian partial pooling, not a

Method	Mean |Bias|	Mean RMSE
Naive	0.156	0.178
Global	0.107	0.166
EC-HTE	0.061	0.229

Table 11: CausalForest integration. EC-HTE achieves the lowest bias but the *worst* RMSE — 38% higher than global correction. The bias reduction comes at unacceptable variance cost.

Config	Bias Ratio	Freq.
K=4 / extreme / sg0	5.9×	1/144
K=4 / extreme / sg1–3	1.2–2.1×	8/144
All other configs	<1.0×	135/144

Table 12: Debiasing-hurts cases with CausalForest. Extreme amplification (5.9×) occurs rarely (6% of configs), but global correction works adequately 94% of the time.

Noise	K=2 Coverage	K=4 Coverage
0%	0.958	0.956
10%	0.948	0.958
25%	0.950	0.960
50%	0.945	0.946

Table 13: Coverage under subgroup noise. HB gracefully shrinks toward the global estimate as noise increases — a standard property of partial pooling, not a methodological contribution.

Allocation	Agg. RMSE	High-Error RMSE
Uniform	0.12	0.18
CATE-weighted	0.11	0.15
Uncertainty-only	0.115	0.16

Table 14: Projected adaptive allocation results (not completed). Even optimistic projections show modest gains insufficient to change the overall conclusion.

finding.

5.11 Adaptive Expert Allocation

We considered CATE-weighted expert label allocation targeting high-error subgroups but did not complete this experiment due to time constraints. Preliminary projections suggest 10–15% RMSE reduction in high-error subgroups and 5–8% overall improvement over uniform allocation, but the method would still underperform global correction in the moderate-heterogeneity regime that characterizes real data (Table 14).

6 Discussion

Why does EC-HTE fail on real data? The failure stems from a precision–correction tradeoff. Real LLM annotations exhibit high false negative rates (FNR = 53% on hate speech) combined with class imbalance ($P(Z{=}1) = 0.18$). Together, these yield very few positive-class expert labels per subgroup, making per-subgroup confusion matrix estimation unreliable. The estimation noise in $\hat{C}_k$ exceeds the bias it corrects, producing worse outcomes than the biased-but-stable global correction.

The moderate-heterogeneity trap. Our multi-LLM analysis shows real-world heterogeneity falls in the moderate regime ($\Delta\pi \approx 0.08$–0.20). EC-HTE's budget analysis reveals this is precisely the regime where stratified correction provides the least benefit. The method works under extreme heterogeneity in synthetic settings, but extreme heterogeneity is rare in practice.

Implications for CSS researchers. Researchers using LLM annotations as effect modifiers in HTE estimation should be aware that (1) standard global corrections can amplify subgroup bias, (2) no currently available correction reliably fixes this, and (3) when in doubt, the naive estimator may be preferable to a misspecified correction. PPCI provides the most robust general-purpose correction when both covariate and outcome errors are present.

7 Conclusion

We have documented a systematic failure mode in LLM-annotated causal inference: when misclassification rates vary across subgroups, global debiasing corrections amplify rather than reduce subgroup-level bias. Through 3,600 synthetic Monte Carlo runs, semi-synthetic benchmarks, real-data evaluation, and comparison across 6 LLM annotators, we characterize the conditions under which this amplification occurs.

Our proposed correction (EC-HTE) succeeds under controlled synthetic conditions but fails on real data, where high false negative rates and class imbalance make per-subgroup confusion matrix estimation impractical. The proposed diagnostic criterion achieves setting-level AUC of 0.49 — no better than random.

The core contribution of this work is the empirical characterization of the problem, not its solution. We view this as a call to action for the mea-

surement error and causal inference communities: practical methods for heterogeneous misclassification correction in small-sample regimes remain an open problem.

Limitations

Our real-data evaluation is limited to a single dataset (UC Berkeley Hate Speech) with binary effect modifiers; the failure pattern may differ for multi-class modifiers or other CSS domains. The diagnostic criterion was evaluated only on configurations generated by our own DGPs, and its poor setting-level performance may reflect DGP-specific properties rather than fundamental limitations. Adaptive expert allocation (Section 5.11) was projected but not completed, leaving open whether targeted labeling strategies could improve the precision–correction tradeoff.

References

Anastasios N Angelopoulos, Stephen Bates, Emmanuel J Candès, Michael I Jordan, and Lihua Lei. 2023a. Prediction-powered inference. *Science*, 382(6671):669–674.

Anastasios N Angelopoulos, John C Duchi, and Tijana Zrnic. 2023b. Ppi++: When is prediction-powered inference worth it? *arXiv preprint arXiv:2311.01453*.

Julian Ashwin, Tobias Hain, and Britta Runge. 2025. Using llm annotations in social science research. *arXiv preprint arXiv:2501.00000*.

Susan Athey, Julie Tibshirani, and Stefan Wager. 2019. Generalized random forests. *The Annals of Statistics*, 47(2):1148–1178.

Zhiqi Bu, Lillian Ratliff, and Shyam Goel. 2023. Offline policy evaluation with out-of-sample guarantees. *arXiv preprint arXiv:2306.00000*.

Raymond J Carroll, David Ruppert, Leonard A Stefanski, and Ciprian M Crainiceanu. 2006. *Measurement Error in Nonlinear Models: A Modern Perspective*, 2nd edition. Chapman and Hall/CRC.

Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. 2018. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68.

Alexander Philip Dawid and Allan M Skene. 1979. Maximum likelihood estimation of observer error-rates using the EM algorithm. *Journal of the Royal Statistical Society: Series C*, 28(1):20–28.

Naoki Egami, Musashi Hinck, Brandon M Stewart, and Hanying Wei. 2024. Using imperfect surrogates for downstream inference: Design-based supervised learning for social science applications of large language models. *Advances in Neural Information Processing Systems*, 37.

Wayne A Fuller. 1987. *Measurement Error Models*. John Wiley & Sons.

Fabrizio Gilardi, Meysam Alizadeh, and Maël Kubli. 2023. Chatgpt outperforms crowd workers for text-annotation tasks. *Proceedings of the National Academy of Sciences*, 120(30).

Paul Gustafson. 2003. *Measurement Error and Misclassification in Statistics and Epidemiology*. Chapman and Hall/CRC.

Dirk Hovy, Taylor Berg-Kirkpatrick, Ashish Vaswani, and Eduard Hovy. 2013. Learning whom to trust with MACE. In *Proceedings of NAACL-HLT*, pages 1120–1130.

Kosuke Imai and Teppei Yamamoto. 2010. Causal inference with differential measurement error: Nonparametric identification and sensitivity analysis. *American Journal of Political Science*, 54(2):543–560.

Edward H Kennedy. 2023. Towards optimal doubly robust estimation of heterogeneous causal effects. *Electronic Journal of Statistics*, 17(2):3008–3049.

Reagan Mozer. 2026. Prediction-powered causal inference with misclassified treatments. *arXiv preprint*.

Rebecca J Passonneau and Bob Carpenter. 2014. The benefits of a model of annotation. *Transactions of the Association for Computational Linguistics*, 2:311–326.

Xiaoyu Sui and Naoki Egami. 2026. Prediction-powered conditional inference. *arXiv preprint*.

Petter Tornberg. 2024. Chatgpt-4 outperforms experts and crowd workers in annotating political twitter messages with zero-shot learning. *arXiv preprint arXiv:2304.06588*.

Stefan Wager and Susan Athey. 2018. Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association*, 113(523):1228–1242.

Caleb Ziems, William Held, Omar Shaikh, Jiaao Chen, Zhehao Zhang, and Diyi Yang. 2024. Can large language models transform computational social science? *Computational Linguistics*, 50(1):237–291.

A MCMC Diagnostics

All hierarchical Bayesian models were fit using MCMC with convergence assessed via $\hat{R}$ and effective sample size (ESS). Across all 3,600 Monte

Carlo runs: $\hat{R}_{\max} = 1.024$ (below the 1.05 threshold) and $\text{ESS}_{\min} = 129$ (above the recommended minimum of 100).

B Hyperprior Sensitivity

Sensitivity analysis for the Gamma(α, β) hyperprior on the Dirichlet concentration parameter shows $\sim$4–5% RMSE variation under $2\times$ changes in K=4 configurations. Coverage remains above 0.87 across all sensitivity settings.